

Wasserstein Kmeans Algorithm

Implementing clustering algorithm over probability density functions

Mathematical Overview

The purpose of this algorithm is to find an optimal way to cluster N PDFs (probability density functions) using the second Wasserstein distance function, defined formally as:

$$W_2^2(P, Q) = \inf_{\gamma \in \Gamma(P, Q)} \int_{\mathbb{R} \times \mathbb{R}} |x - y|^2 d\gamma(x, y)$$

where P and Q are PDFs, and $\Gamma(P, Q)$ is the set of joint distributions with marginal density functions P and Q .

We know from (Le Gouic and Loubes 2016) that if P and Q are univariate:

$$W_2^2(P, Q) = \int_0^1 (F_P^{-1}(u) - F_Q^{-1}(u))^2 du$$

where $F_X^{-1}(u)$ is the inverse CDF (cumulative distribution function) for PDF X .

The task of finding a centroid for N data points can be achieved by finding Y such that:

$$Y = \arg \min_{Y \in B} \sum_{i=1}^N d(X_i, Y)$$

where our data points exist in set $S \subseteq B$, and d is an arbitrary distance function.

Let $d = W_2^2$ (the squared Wasserstein distance). Then we wish to find a point Y such that:

$$Y = \arg \min_{Y \in B} \sum_{i=1}^N \int_0^1 (F_Y^{-1}(u) - F_{X_i}^{-1}(u))^2 dt$$

From (Le Gouic and Loubes 2016) we know that:

$$F_Y^{-1} = \frac{1}{N} \sum_{i=1}^N F_{X_i}^{-1}(u)$$

Thus, we can easily find a the best fit cluster of our PDFs by averaging the quantile functions of our data points. If we wanted to assign each point to a set of K clusters, the objective is to find a sequence of k clusters $\mu_1, \dots, \mu_K$:

$$\mu_1, \dots, \mu_K = \arg \min_{\mu_1, \dots, \mu_K} \sum_{k=1}^K \sum_{X_i \in C_k} d(X_i, \mu_k)$$

We can find a local minimum to this problem if, at step $t + 1$, we update each cluster using update procedure:

$$\mu_k^{t+1} \leftarrow \frac{1}{|C_k^t|} \sum_{X_i \in C_k^t} F_{X_i}^{-1}(u)$$

Sources

Le Gouic, Thibaut, and Jean-Michel Loubes. 2016. “Existence and Consistency of Wasserstein Barycenters.” *arXiv Preprint arXiv:1506.04153*. <https://arxiv.org/abs/1506.04153>.